

B.Sc. Semester-V 2023
Computer Science
CC-12: Theory of Computation

Time: 3 hours

Full Marks: 60

Answer question number 1 and any five from the rest

- 1 a) Find the language generated by the grammar: $S \rightarrow OA \mid IS \mid 0 \mid 1, 011, A \rightarrow IA \mid IS \mid 1$.
b) Construct a DFA accepting all strings over $\{a, b\}$ ending in ab .
c) When do you say that a grammar is in Chomsky Normal Form?
d) What do you mean by the yield of a derivation?
e) When can we say that two finite automata over Σ are equivalent? [2×5]

- 2 a) Construct a Moore machine which is equivalent to the following Mealy machine:

Present State	Next State			
	$a=0$		$a=1$	
	State	Output	State	Output
q_1	q_1	1	q_2	0
q_2	q_4	1	q_4	1
q_3	q_2	1	q_3	1
q_4	q_3	0	q_1	1

- b) Write the algorithm for the construction of minimum automaton. [5+5]
- 3 a) If the grammar G is given by the productions $S \rightarrow aSa \mid bSb \mid aa \mid bb \mid \Lambda$, show that
(i) any string in $L(G)$ is of length $2n, n \geq 0$
(ii) the number of strings of length $2n$ is 2^n .
b) Consider the grammar G given by $S \rightarrow OSA2, S \rightarrow 012, 2A \rightarrow A2, 1A \rightarrow 11$. Test [6+4]
whether (a) $00112 \in L(G)$ and (b) $001122 \in L(G)$.
- 4 a) Construct a DFA accepting all strings H over $\{0,1\}$ such that the number of 1's in H is 3 mod 4.
b) Let L be the set of all palindromes over $\{a, b\}$. Construct a grammar G generating L . [4+6]
- 5 a) State and prove Arden's theorem.
b) Show that the set $L = \{a^{i^2} \mid i \geq 1\}$ is not regular. [5+5]
- 6 a) Find (i) a leftmost derivation, (ii) a rightmost derivation, and (iii) a derivation which is neither leftmost nor rightmost of $abababa$, given that $abababa$ is in $L(G)$, where G is the grammar given by $S \rightarrow SbS \mid a$.
b) Reduce the following grammar to CNF: $S \rightarrow abSb \mid a \mid aAb, A \rightarrow bS \mid aAAb$ [5+5]
- 7 a) Define (i) ID of a PDA (ii) IID of a PDA and (iii) moves in a PDA
b) Consider the transition table for a Turing machine as follows and obtain the computation sequence of 000 to declare if it is accepted or not. [6+4]
- | Present State | 0 |
|--------------------|---------|
| $\rightarrow(q_1)$ | bq_1R |
| q_2 | bq_2R |
- 8 Write short notes on:
a) Chomsky's classification of languages
b) Removal of Unit productions. [5+5]